Congratulations! You passed!

TO PASS 80% or higher

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100%

Practice Assessment

TOTAL POINTS 10

1. Which of the following are the most accurate characterizations of sample models and distribution models? (Select all that apply) 1 Point(s)

- [x] Both sample models and distribution models can be used to obtain a possible next state and reward, given the current state and action.

  Correct
  Correct, given any state and action, you can sample the next state and reward using a sample model or distribution model.

- [ ] A sample model can be used to compute the probability of all possible trajectories in an episodic task based on the current state and action.

- [x] A distribution model can be used as a sample model.

  Correct
  Correct, a distribution model contains all the information about the transition dynamics of the system, which can be used to "sample" new states and rewards given the current state and action. I can like a sample model.

- [ ] A sample model can be used to obtain a possible next state and reward given the current state and action, whereas a distribution model can only be used to compute the probability of this next state and reward given the current state and action.

2. Which of the following statements are TRUE for Dyna architecture? (Select all that apply) 1 Point(s)

- [x] Simulated experience can be used to improve the value function and policy.

  Correct
  Correct, we do this in the planning step of the tabular Dyna-Q algorithm.

- [x] Real experience can be used to improve the model.

  Correct

- [ ] Simulated experience can be used to improve the model.

- [x] Real experience can be used to improve the value function and policy.

  Correct
  Correct, we do this in the direct-RL step of the tabular Dyna-Q algorithm.

3. Mark all the statements that are TRUE for the tabular Dyna-Q algorithm. (Select all that apply) 1 Point(s)

- [x] For a given state-action pair, the model predicts the next state and reward

  Correct
  Correct, this is because in the tabular Dyna-Q algorithm, the model stores the next state and action for every state-action pair that is encountered.

- [ ] The memory requirements for the model in case of a deterministic environment are quadratic in the number of states.

  Correct
  Correct, the algorithm assumes that the environment deterministically transitions to a single next state for every state-action pair. Therefore, the memory requirements for the tabular Dyna-Q model scale only in proportion with the number of states and actions, and not with the number of next state and reward transitions. So when the update model step is modified, we would be losing a lot of information. This may lead to a poor performance even though we are using a planning based method.

- [ ] The algorithm **cannot** be extended to stochastic environments.

4. Which of the following statements are TRUE? (Select all the apply) 1 Point(s)

- [x] Model-based methods like Dyna typically require more memory than model-free methods like Q-learning.

  Correct
  Correct, additional memory is required to store the model.

- [x] When compared with model-free methods, model-based methods are relatively more varied.

  Correct
  Correct, we have seen examples of this in the lectures and Chapter 8 of Sutton and Barto's RL textbook.

- [x] The amount of computation per interaction with the environment is large in the Dyna-Q algorithm (with non-zero planning steps) as compared to the Q-learning algorithm.

  Correct
  Correct, apart from the direct-RL steps performed in the Q-learning algorithm, Dyna-Q performs additional steps of model-learning and planning.

- [x] Model-based methods often suffer more from bias than model-free methods, because of inaccuracies in the model.

  Correct
  Correct, the performance of model-based methods depends heavily on the model.

5. Which of the following is generally the most computationally expensive step of the Dyna-Q algorithm? 1 Point(s)

Tabular Dyna-Q

Initialize Q(s, a) and Model(s, a) for all $s \in S$ and $a \in A(s)$
Loop forever:
(a) $S \leftarrow$ current (nonterminal) state
(b) $A \leftarrow \epsilon$-greedy(S, Q)
(c) Take action A, observe resultant reward, R, and state, S'
(d) $Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma \max_a Q(S', a) - Q(S, A)]$
(e) $Model(S, A) \leftarrow R, S'$ (assuming deterministic environment)
(f) Loop repeat n times:
$S \leftarrow$ random previously observed state
$A \leftarrow$ random action previously taken in S
$R, S' \leftarrow Model(S, A)$
$Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma \max_a Q(S', a) - Q(S, A)]$

- [ ] Model learning (step e)

- [ ] Direct-RL (step d)

- [ ] Action selection (step b)

- [x] Planning (indirect-RL, step f)

  Correct
  Correct, the planning step performs search control (S,A) with an exponential Monte-Carlo implementation, generates simulated experience (S,A,R), and updates the action-value function (S,A,R). This is repeated N times, for overall O(N*|A|) time complexity.

6. What are some possible reasons for a model-learning learned by a model-based method to be inaccurate? (Select all that apply) 1 Point(s)

- [ ] The agent's policy has changed significantly from the beginning of training.

- [ ] There is too much exploration (e.g., epsilon is greater greedy exploration is set to a high value of 0.5)

- [x] The environment has changed.

  Correct
  Correct, if the environment has changed (e.g., a new wall has come up in the gridworld), changing the transition probabilities, then the learned model is no longer accurate.

- [x] The transition dynamics of the environment are stochastic, and only a few transitions have been experienced.

  every sample is form reliable estimate in the model. For a stochastic environment, we can keep counts of the number of times each next state and reward is experienced from each state-action pair. We can use this to estimate probabilities of next states and rewards, from a given state and action.

7. In search control, which of the following methods is likely to make a Dyna agent perform better in the early episodes of the environment? Assume that the environment is a gridworld with a goal location and that the search control is the process that selects the starting states and actions in planning. Also recall the navigation example in the video lectures in which a large number of transitions from the start state were added to the of the basic search control procedure in the Dyna-Q algorithm. (Select the best option) 1 Point(s)

- [ ] Select state-action pairs uniformly at random from all previously experienced pairs.

- [x] Start backsearch from state-action pairs that have had a non-zero update (e.g. from the state right beside a goal state). This avoids the otherwise wasteful computations from state-action pairs which have had no updates.

- [ ] Start with state-action pairs encountered in a fixed order (e.g., in a gridworld, state top-left or bottom-right, actions up, down, left, right)

- [ ] All of these are equally good/bad.

  the intuition behind backward focusing and prioritized sweeping (check out Section 8.4 of Sutton and Barto's RL textbook).

8. In the lectures, we saw how the Dyna-Q+ agent found the newly opened shortcut in the shortcut maze, whereas the Dyna-Q agent didn't. Which of the following implications drawn from the figure are TRUE? (Select all that apply) 1 Point(s)

- [x] The Dyna-Q+ agent performs better than the Dyna-Q agent even in the first half of the experiment because of the increased exploration.

  Correct
  Correct, the increased exploration due to the reward bonus helps the agent discover the path to the goal relatively faster.

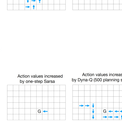
- [ ] The Dyna-Q agent can never discover shortcuts (i.e., when the environment changes to become better than it was before).

- [x] The difference between Dyna-Q+ and Dyna-Q increased slightly over the first part of the experiment. This is because the Dyna-Q+ agent keeps exploring even when the environment isn't changing.

  Correct
  Correct, such exploration can lead to a slightly suboptimal behaviour even if the optimal policy has been learned for a stationary environment.

- [ ] None of the above are true.

of the episode, and all rewards were zero during this episode except for a positive reward at G.



Now which of the following figures best depicts the action values that would be increased by the end of the episode using one-step Sarsa and 100-step planning Dyna-Q? (Select the best option)

- [ ]

- [ ]

- [x]

- [ ]

9. Expected Sarsa 1 Point(s)

- [ ]

10. Which of the following are planning methods? (Select all that apply) 1 Point(s)

- [x] Value Iteration

  Correct
  Correct, Value Iteration is a Dynamic Programming method that uses a model to improve the policy.

- [x] Dyna-Q

  Correct
  Correct, Dyna-Q combines model-free Q-learning with planning. It uses both the experience from the environment as well as simulated experiences from the model in order to make updates to improve the policy.

- [ ] Expected Sarsa

- [ ] Monte-Carlo